CHAPTER 39

Development of the ear

INNER EAR

The production of a precisely positioned and functionally well-tuned inner ear depends on genetic patterning and a cascade of transcription signals expressed by numerous tissues, including the developing inner ear and its surrounding periotic mesenchyme, the adjacent hindbrain, neural crest and notochord (Ohyama et al 2010, Sienknecht 2013).

The first signs of inner ear development are visible shortly after those associated with the developing eyes. Two patches of ectodermal thickening, the otic placodes, appear lateral to the hindbrain at stage 9. Each placode invaginates as an otic pit, adjacent to rhombomeres 5 and 6 of the hindbrain and dorsal to the second pharyngeal cleft. During stage 12, the pit is pinched off from the surface ectoderm to form a simple, hollow epithelial sac, the otic vesicle (otocyst or auditory vesicle) (Fig. 39.1).

Regions of the vesicle differentiate into prosensory domains, which give rise to the membranous labyrinth and the vestibulocochlear (statoacoustic) ganglia of the eighth cranial nerve. The first morphological evidence of this differentiation is visible during stage 14 (approximately 33 days), when the otic vesicle loses its initial piriform shape. A tubular diverticulum, the endolymphatic appendage, develops from its dorsomedial rim. The remainder of the vesicle, the utriculosaccular chamber, differentiates into an expanded pars superior and a narrower pars inferior. The endolymphatic appendage elongates and its tip expands into an endolymphatic sac that is connected to the pars superior by a narrow endolymphatic duct.

Two plate-like diverticula, one vertical and one horizontal, emerge from the dorsal part of the pars superior. The epithelia in the centre of each outgrowth coalesce to form a fusion plate; the central part of this plate is eventually resorbed, leaving the anlagen of the semicircular canals. The vertical plate gives rise to the anterior and posterior semicircular canals, which share a common crural attachment to the utriculosaccular chamber, and the horizontal plate gives rise to the lateral semicircular canal. A small expansion, the ampulla, forms at one end of each semicircular canal.

The central part of the utriculosaccular chamber, which now represents the membranous vestibule, becomes divided into a small ventral saccule and a larger dorsal utricle, mainly by horizontal infolding extending from the lateral wall of the chamber towards the opening of the endolymphatic duct. In this way, communication between the utricle and saccule is restricted to a narrow utriculosaccular duct. The latter becomes acutely bent on itself; its apex is continuous with the endolymphatic duct. While this is happening, the membranous labyrinth rotates so that its long axis, which was originally vertical, becomes more or less horizontal.

The ventral tip of the pars inferior begins to elongate. A medially directed evagination, the cochlear anlagen, is evident in the ventral part

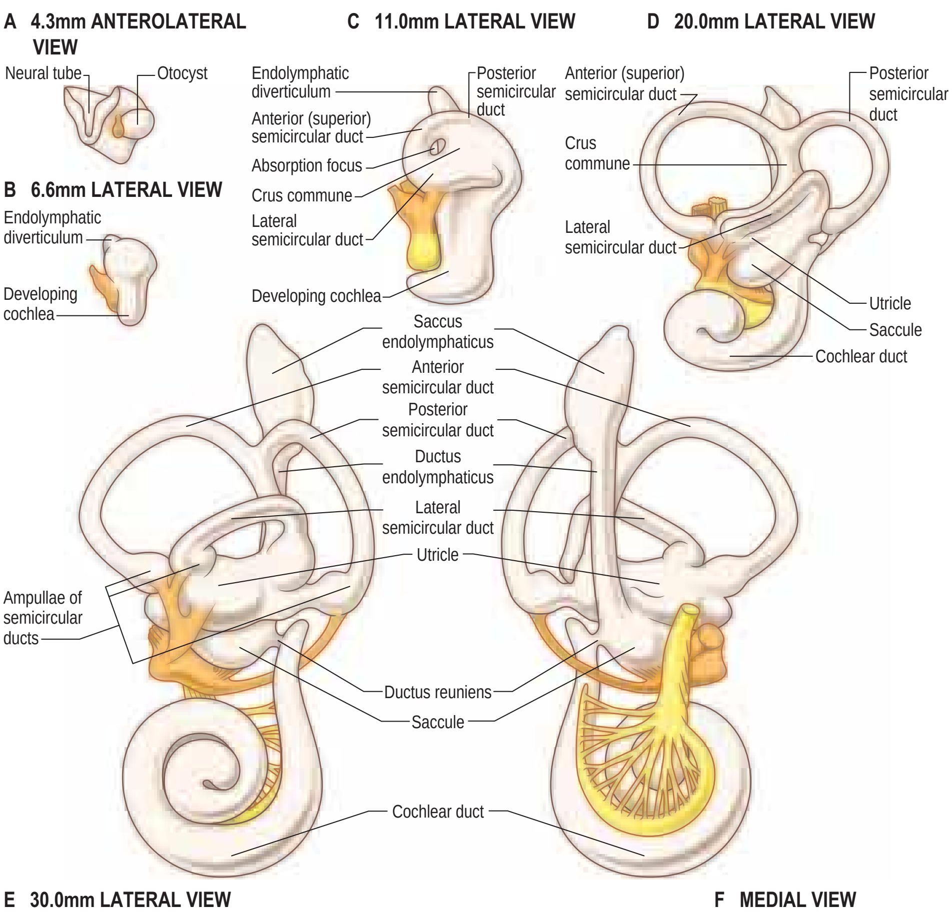

Fig. 39.1 A–F, The stages in the development of the membranous labyrinth from the otocyst, at the embryonic stages and aspects indicated. Note the relationship of the vestibular (orange) and cochlear (yellow) parts of the vestibulocochlear nerve.

of the utriculosaccular chamber in a 7–9 mm (approximately 35 days) embryo. The proximal region of this cochlear duct continues to increase in length and its distal region becomes progressively more coiled. When the duct has achieved its final length and spiral configuration, its proximal part becomes constricted, forming the ductus reuniens by which the saccule remains connected to the cochlea. The cochlea is initially patterned into several prosensory domains. The central domain will give rise to the organ of Corti. Molecular signals regulating the induction and differentiation of the organ of Corti are complex (Kelly and Chen 2009). Relatively little is known about the mechanisms establishing sensory and non-sensory territories in the cochlear duct; there is evidence that bone morphogenetic protein (BMP) signalling is required (Ohyama et al 2010).

Cells derived from the otocyst differentiate into the bipolar neurones that populate the vestibular and cochlear ganglia; sustentacular cells (Wan et al 2013); the unique endolymph-producing epithelia of the stria vascularis; the absorbing epithelia of the endolymphatic sac; the general epithelial lining of the membranous labyrinth; and specialized cells in the six sensory patches of the inner ear (crista ampullaris of the three semicircular canals, maculae of the utricle and saccule, organ of Corti in the cochlea). Each of the sensory patches consists of mechanosensory hair cells and non-sensory supporting cells arranged into mosaic patterns that are essential for normal hearing and balance. The only cells in the mature inner ear that are not of otocyst origin are melanocytes in the stria vascularis, which are derived from the neural crest.

The very different morphologies of the mature cristae, maculae and organ of Corti reflect the differential expression of multiple genes during their development and maturation. The formation of the utricle and saccule occurs during a time that coincides with the initiation of hair cell planar polarity. It is important to note that the development of core planar cell polarity in the orientation of stereocilia in the mechanosensory hair cells of the mammalian inner ear is the subject of an extensive and conflicting literature (Deans 2013, Ezan and Montcouquiol 2013).

At the same time as these changes are taking place, mesenchymal cells surrounding the developing membranous labyrinth chondrify to form an encasing cartilaginous otic capsule. Between 16 and 23 weeks, the otic capsule ossifies to form all of the bony labyrinth of the internal ear within the petrous temporal bone, except the modiolus and osseous spiral lamina, which ossify directly from connective tissue. The cartilaginous capsule is initially incomplete and the cochlear, vestibular and facial ganglia are temporarily exposed in the gap between its canalicular and cochlear parts. They subsequently become covered by an outgrowth of cartilage, in which the facial nerve becomes enclosed.

Vacuoles filled with perilymph develop in the embryonic connective tissue between the cartilaginous capsule and the epithelial wall of the membranous labyrinth. The rudiment of the periotic cistern or vestibular perilymphatic space can be seen in the reticulum between the saccule and fenestra vestibuli in an 11–14 mm (approximately 42 days) embryo. The cochlear scalae (scala vestibuli and scala tympani) develop by fusion of these spaces; the mechanism is uncertain (Kim et al 2011). The two scalae gradually extend along each side of the cochlear duct; a communication, the helicotrema, opens between them when they reach the tip of the coiled duct.

The rudiment of the eighth cranial nerve appears in the 4–6 mm embryo as the vestibulocochlear (statoacoustic) ganglion that lies between the otocyst and the hindbrain and is initially temporarily fused with the ganglion of the facial nerve. Neuroblasts, whose fate is already specified in the ectoderm of the otic placode, delaminate from the anteroventral region of the otic vesicle, proliferate and migrate to the site of the presumptive ganglion, where they coalesce. They differentiate into mature bipolar sensory neurones and become segregated into vestibular and spiral ganglia, each associated with the corresponding division of the eighth cranial nerve. These neurones are unusual because they are exclusively placodal in origin, unlike neurones in most cranial ganglia (which have a dual placodal and neural crest origin), and many of their somata become enveloped in thin myelin sheaths. Their peripheral processes collectively provide the afferent innervation of the labyrinthine hair cells, relaying balance and auditory information. Tantalizing evidence about the transcriptional networks that encode the positioning of sensory hair cells and spiral ganglion neurones along a frequency (tonotopic) gradient within the developing cochlea, and which ensure that the two cell types make appropriate functional connections, is appearing in animal models (Appler and Goodrich 2011, Appler et al 2013, Coate and Kelley 2013). The olivocochlear bundle, an outgrowth of axons from neurones in the superior olivary complexes in the pons, accompanies the axons of the developing eighth cranial nerve; these axons provide the inner ear with an efferent innervation, mainly to the outer hair cells in the organ of Corti, where they are associated with modulation of hearing. *In utero*, the fetus receives sound by bone conduction; interaural sound differences are not established until birth, whether preterm or term.

MIDDLE EAR (TYMPANIC CAVITY AND PHARYNGOTYMPANIC (AUDITORY) TUBE)

All the components of the outer and middle ears develop from the first and second pharyngeal arches. The pharyngotympanic tube and tympanic cavity are lateral extensions of the early pharynx (see Fig. 36.4). They become visible in a 4–6 mm embryo as a hollow, the tubotympanic recess, lying between the first and second pharyngeal arches, with a floor that consists of the second arch and its limiting pouches. The forward growth of the third pharyngeal arch causes the proximal part of the recess to remain narrow, forming the pharyngotympanic tube region, and also excludes the inner part of the second arch from this portion of the floor. The more lateral part of the recess eventually comes in contact with the first pharyngeal cleft and widens and develops into the tympanic cavity; its floor later forms the lateral wall of the tympanic cavity up to approximately the level at which the chorda tympani branches off from the facial nerve. The lateral wall of the tympanic cavity contains first and second arch elements. The first arch territory is limited to that part in front of the anterior process of the malleus, and the second arch forms the outer wall behind this and also turns on to the posterior wall to include the tympanohyal region.

The tubotympanic recess at first lies inferolateral to the cartilaginous otic capsule; this spatial relationship alters as the capsule enlarges and the tympanic cavity becomes anterolateral. A cartilaginous process grows from the lateral part of the capsule to form the tegmen tympani (Rodríguez-Vázquez et al 2011). The process curves caudally to form the lateral wall of the pharyngotympanic tube, incorporating the tympanic cavity and the proximal part of the pharyngotympanic tube into the petrous region of the temporal bone. The mastoid antrum appears as a dorsal expansion of the tympanic cavity during the sixth to seventh months (some sources place this as a later phenomenon). Exactly how middle ear cavitation occurs is poorly understood (Sienknecht 2013); it seems to involve cavitation of a neural crest mass (Thompson and Tucker 2013).

The middle ear ossicles are of neural crest origin, i.e. crest cells that have migrated from rhombomeres 1–4 into the mesenchyme of the first and second pharyngeal arches. The malleus develops from the dorsal end of the ventral mandibular (Meckel's) cartilage of the first arch. The incus develops from the dorsal cartilage of the first arch, which is probably homologous to the quadrate bone of birds and reptiles. The origin of the stapes in humans remains controversial. It is thought to be derived mainly from an anlage situated in the cranial end of the cartilage of the second pharyngeal arch, initially as a ring (anulus stapes) that encircles the small stapedial artery (Rodríguez-Vázquez 2005). The ossicles remain embedded in the mesenchymal roof of the tympanic cavity until the eighth month of gestation, when the mesenchyme is resorbed. As this happens, the ossicles become suspended within the developing tympanic cavity, initially by transient endodermal mesenteries and ultimately by supporting ligaments. They become covered by the mucosa of the middle ear as the tympanic cavity fills with air after birth. Further postnatal developmental changes contribute to the functional maturation of the middle ear.

Two anlagen of each stapedius muscle appear close to the stapedial artery and facial nerve in second arch mesenchyme in 13–17 mm embryos. Tensor tympani starts to appear near the extremity of the tubotympanic recess at almost the same time in first arch mesenchyme. The pyramidal eminence is formed within a condensation of mesenchyme around the belly of stapedius (Rodríguez-Vázquez 2009). The pharyngeal membrane separating the tympanic cavity from the external acoustic meatus develops into the tympanic membrane.

EXTERNAL EAR

The external acoustic meatus develops from the dorsal end of the first pharyngeal (hyomandibular) cleft (see Fig. 36.4). Close to its dorsal extremity, this groove extends inwards as a funnel-shaped primary meatus, from which the entire cartilaginous part of the meatus, and a small area of its roof, are developed. A solid epidermal plug extends inwards from the tube along the floor of the tubotympanic recess. The cells in the centre of the plug subsequently degenerate to produce the inner part of the meatus (secondary meatus). The epidermal stratum of the tympanic membrane is formed from the deepest ectodermal cells of the epidermal plug; the fibrous stratum is formed from the mesenchyme between the meatal plate and the endodermal floor of the tubotympanic recess. The osseous part of the external acoustic meatus develops postnatally from the tympanic ring of the squamous part of the temporal bone; the cartilaginous part develops much earlier and independently of the osseous part (Ikari et al 2013).

The development of the auricle is initiated by the appearance of six tissue elevations, the auricular hillocks, which form round the margins of the dorsal portion of the first pharyngeal cleft. Of the six, three are on the caudal edge of the first pharyngeal (mandibular) arch and three on the cranial edge of the second pharyngeal (hyoid) arch (see Fig. 36.3A). The hillocks appear from stage 15; before then, only the most ventral hillock on the mandibular arch, which subsequently forms the tragus, can be identified. The rest of the auricle is formed in the mesenchyme of the hyoid arch, which extends forwards round the dorsal end of the remains of the first pharyngeal cleft, forming a keel-like elevation that is the forerunner of the helix. The contribution made by the mandibular arch to the auricle is greatest at the end of the second month; thereafter, this contribution becomes relatively reduced as growth continues and, eventually, the area of skin supplied by the mandibular nerve extends little above the tragus. The lobule is the last part of the auricle to develop.

Common congenital anomalies of the auricle are described in Chapter 37.

HEREDITARY DEAFNESS

Prelingual deafness is the most common congenital anomaly, affecting 2–6/1000 newborns: at least two-thirds of cases reported in the developed world are due to genetic factors, classified as non-syndromic (typically sensorineural and accounting for approximately 70% of cases) and syndromic (which may be sensorineural, conductive or mixed and accounts for approximately 30% of cases). A number of molecules likely to be mediators of genetic hearing loss have been identified, including *TGFB1*, *BMP4*, *ERK1*/2 and many *GPCR* genes (Stamatiou and Stankovic 2013). The range of affected chromosomes and the specific genes involved in sensorineural deafness is given by Hildebrand et al (2010). The structural abnormalities resulting from absent or aberrant gene expression during the development of the ear include labyrinthine dysplasias (e.g. a reduction in the number of cochlear turns or an enlargement of the endolymphatic duct); disorganization of the patterns of stereocilia in sensory hair cells and loss of cytoskeletal components in these cells; and defects in K^+ recycling across gap junctions between activated hair cells.

SECTION 4

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